



Quantitative Evaluation of Genetic programming for EFSM Inference

Petrea Norgren

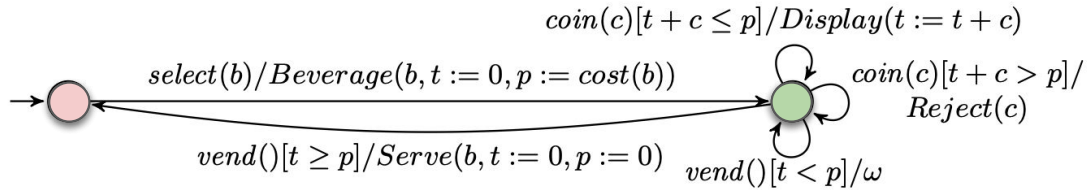
Supervisors: Roland Groz, Catherine Oriat & German Vega

Lab: LIG



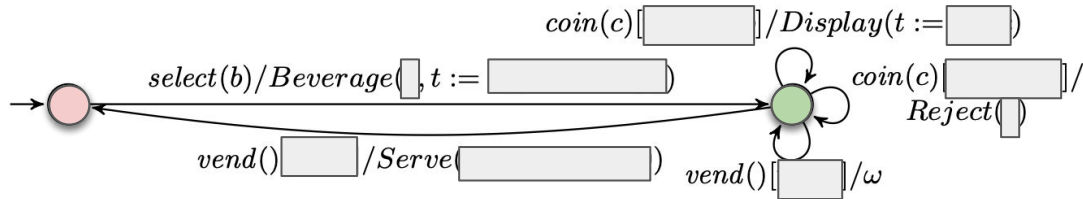
System inference

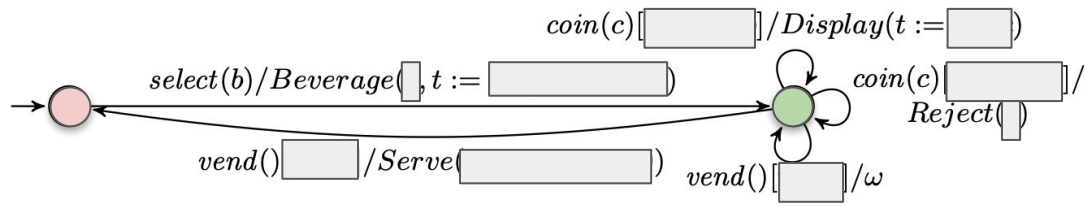
From black box -> Model



System inference

From black box -> Model



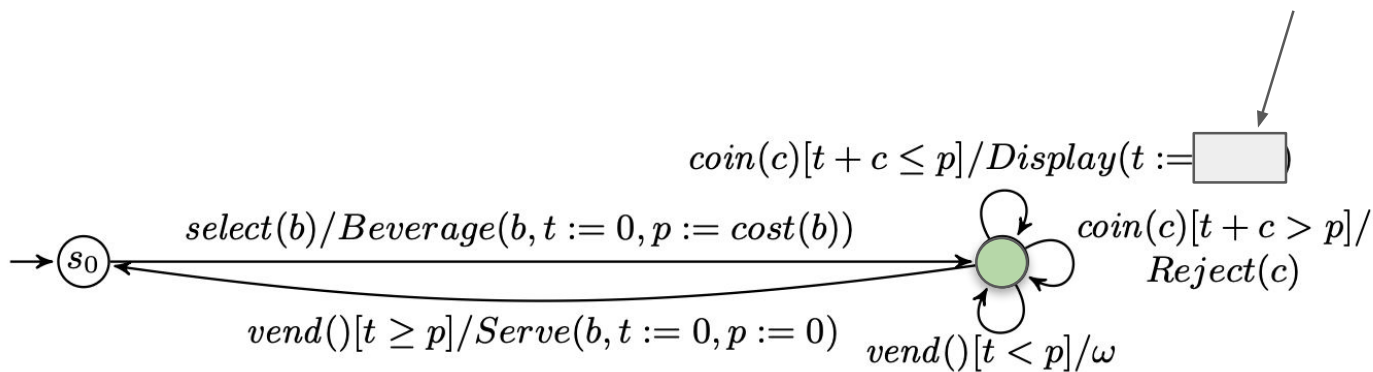


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coffee		select()			Bev.()	0		
	20	coin()	0	cost(b)	Disp.()	20		
	10	coin()	20	cost(b)	Disp.()	30		
	50	coin()	30	cost(b)	Disp.()	80		
	100	coin()	80	cost(b)	Rej.()	80	100	
	20	coin()	80	cost(b)	Disp.()	100		
		vend()	100	cost(b)	Serv.()	0		coffee
			0					



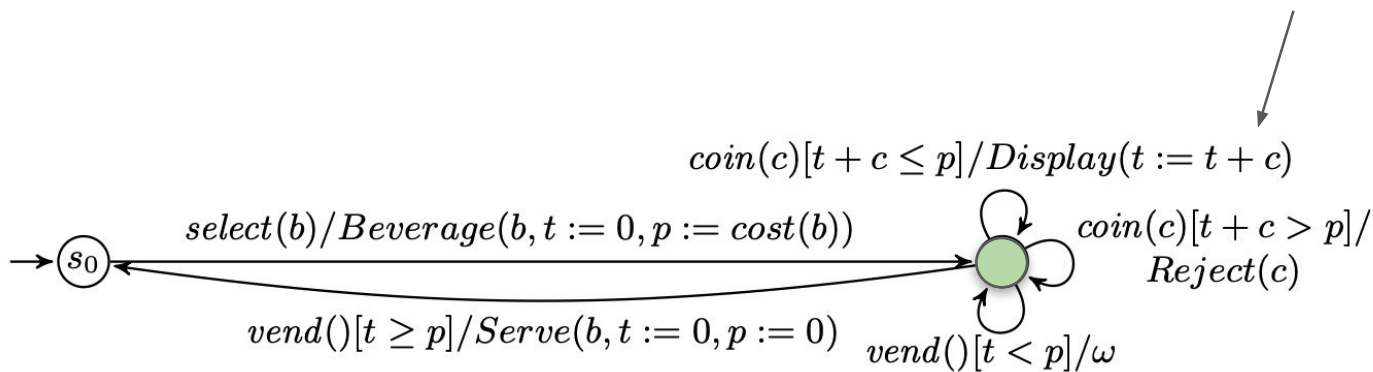
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coffee		select()			Bev.()	0		
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	50	coin()	30	cost(b)	Disp.()	80		
	100	coin()	80	cost(b)	Rej.()	80	100	
	20	coin()	80	cost(b)	Disp.()	100		
		vend()	100	cost(b)	Serv.()	0		coffee
			0					

c	input	t	p	Output	t'
20	coin()	0	cost(b)	Disp.()	20
10	coin()	20	cost(b)	Disp.()	30
50	coin()	30	cost(b)	Disp.()	80
20	coin()	80	cost(b)	Disp.()	100



b	c	input	t	p	Output	t'	c'	b'
coffee		select()			Bev.()	0		
	20	coin()	0	cost(b)	Disp.()	20		
	10	coin()	20	cost(b)	Disp.()	30		
	50	coin()	30	cost(b)	Disp.()	80		
	100	coin()	80	cost(b)	Rej.()	80	100	
	20	coin()	80	cost(b)	Disp.()	100		
		vend()	100	cost(b)	Serv.()	0		coffee
			0					

c	input	t	p	Output	t'
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50	coin()	30	cost(b)	Disp.()	80
20	coin()	80	cost(b)	Disp.()	100



Goals & Purpose

Goal: Create a quantitative evaluation metric for the GP-inference tool

How close to the truth is the inferred function?

Purpose: To help improve the tool by being able to evaluate alterings in the model

Did this altering make the inferred functions better? Worse?

The idea

Can correlations give us an idea of the source of the error?

True function: $2 \cdot i_0 + i_1$

Inferred: $2 \cdot i_0 + i_1 + r_1$

Absolute value
of correlation
coefficients

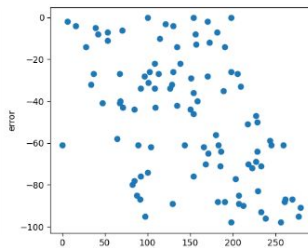
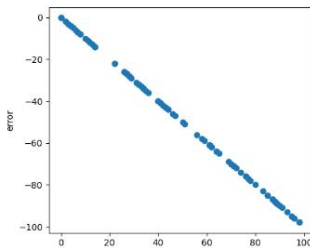
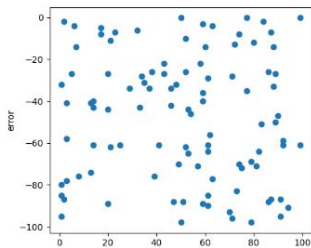
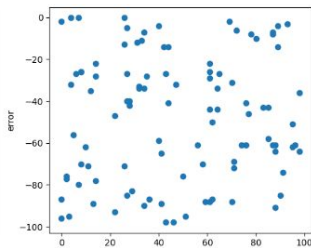
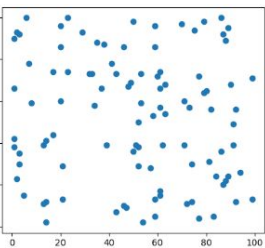
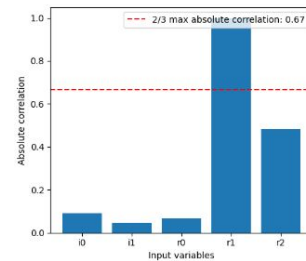
i_0 vs residuals

i_1 vs residuals

r_0 vs residuals

r_1 vs residuals

r_2 vs residuals



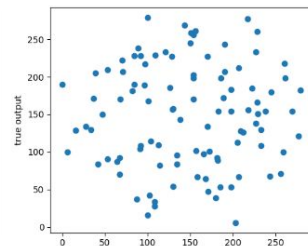
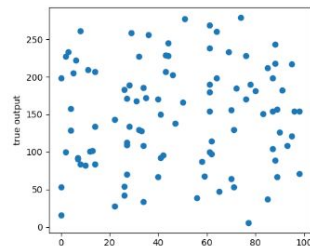
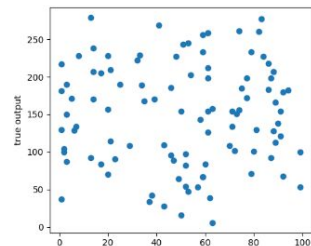
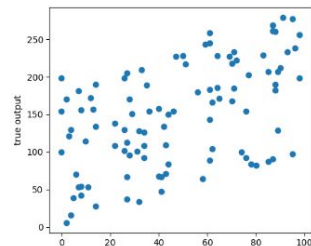
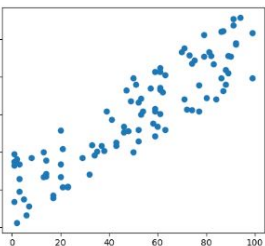
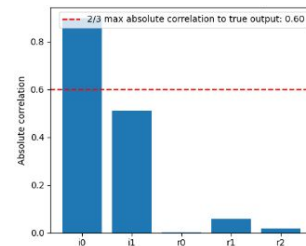
i_0 vs true output

i_1 vs true output

r_0 vs true output

r_1 vs true output

r_2 vs true output



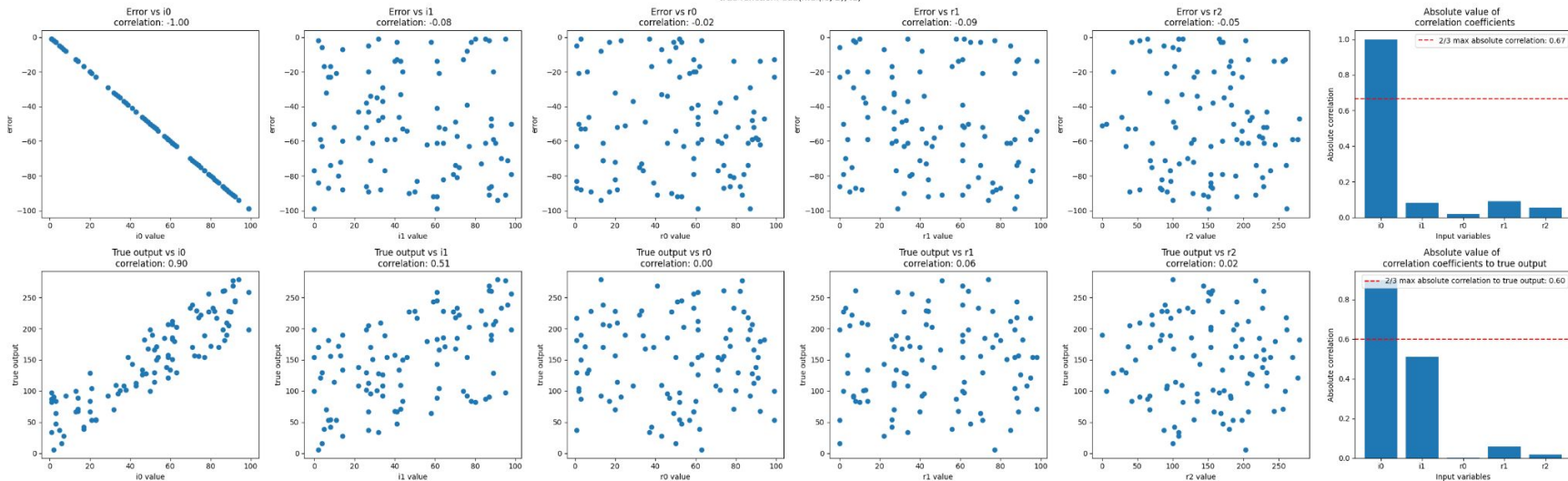
Example

1 spiking correlation

True function: `add(mul(i0, 2), i1)`
 Inferred function: `add(mul(i0, 3), i1)`

	i0	i1	r0	r1	r2	o	inferred
0	51	88	25	61	0	190	241
1	92	59	51	88	190	243	335
2	14	40	92	59	243	68	82
3	71	28	14	40	68	170	241
4	60	14	71	28	170	134	194

Type of error: "off by scale in one variable"
 Inferred function: `add(mul(i0, 3), i1)`
 true function: `add(mul(i0, 2), i1)`



Evaluation - Error categories

01 Inferred function is correct

The inferred function is assumed to be correct until counter examples occur.

02 Constant error

Cause of error: Inferred function has an incorrectly added constant term.

03 One spiking correlation coefficient

Likely cause of error:

- A variable in the inferred function has the wrong coefficient.
- A variable is present in the inferred function and not in the true function.
- A variable is missing from the inferred function that is present in the true function.

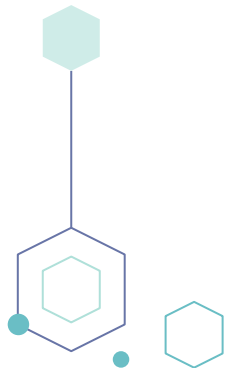
04 > 1 Spiking correlation coefficients

Likely cause of error: A combination of the errors mentioned above or worse.

Evaluation of GP tool

01 Data generation

02 Evaluation



Data generation

01

Inputs

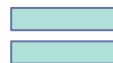
Numpy randint()

Parameters

	i0	i1	r0	r1
0	51	88	25	61
1	92	59	51	88
2	14	40	92	59
3	71	28	14	40
4	60	14	71	28



`add(mul(i0, 2), i1)`



02

Function

Recursive process.

03

Output

Output column and register initialized

	i0	i1	r0	r1	r2	o
0	51	88	25	61	0	190
1	92	59	51	88	190	243
2	14	40	92	59	243	68
3	71	28	14	40	68	170
4	60	14	71	28	170	134

Evaluation of GP tool - set-up

The parameters used for the evaluation:

Functions tested per complexity	3
GP runs per function	5
Generations in GP	100, 200, 300
Table Lengths tested	20, 40, 60
Parameter Value Range	[0,9]
Constant Value Range	[1,10]
Probability of Constant	$1/(\text{\#parameters} + 1)$

Results

- In general errors increase with complexity
- Increasing number of generations has less effect

The results reflect known properties of the GP tool.

This suggests the evaluation works.

Direction of increasing number of generations

Direction of increasing number of input parameters

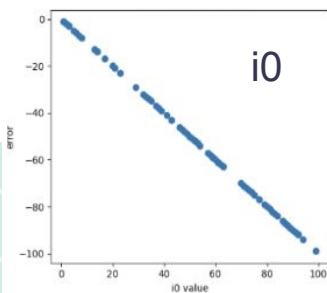


Accurate Inference Constant error One high correlation $\geq$ High Correlation

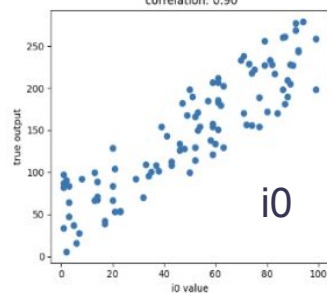
Future work

Implement residual analysis in the GP tool to make better choices among the candidates at each step.

True function: $2*i0+i1$
 Inferred: $3*i0+i1$

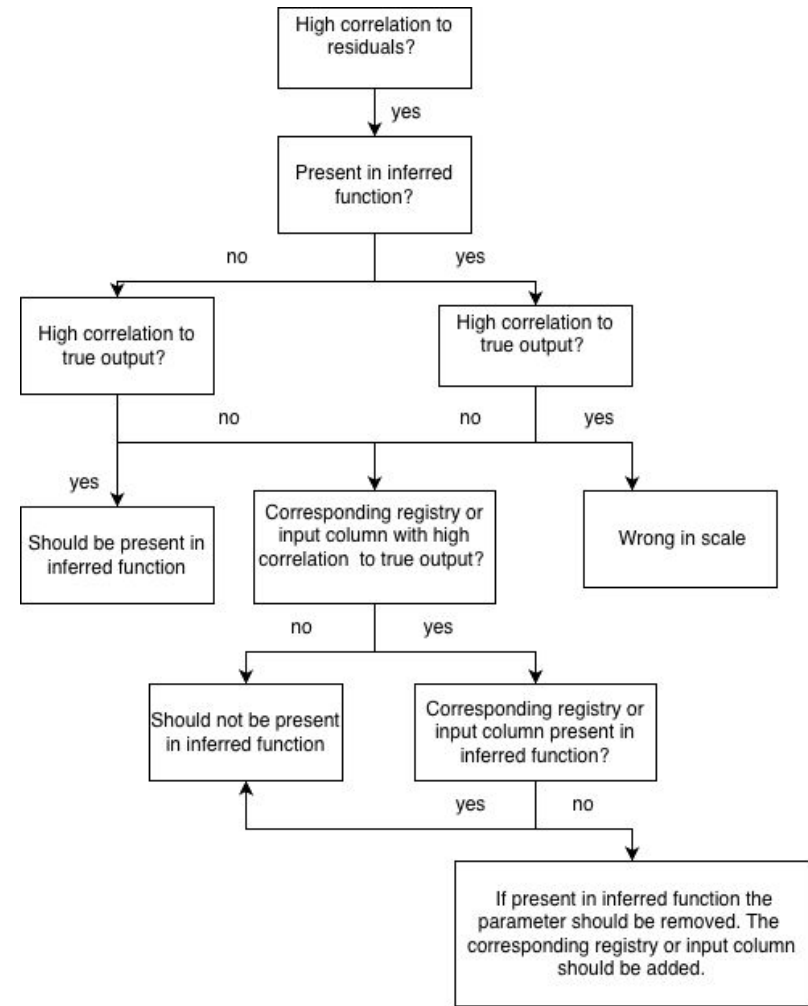
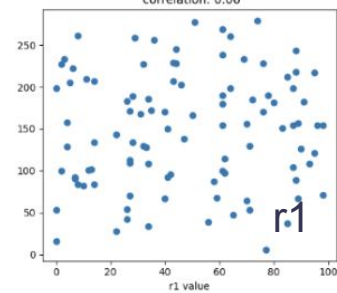
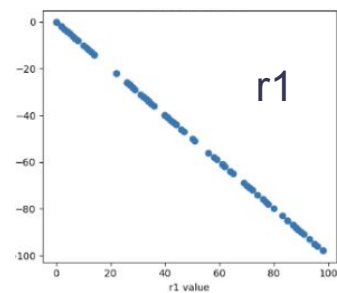


Correlation residuals



Correlation true output

True function: $2*i0+i1$
 Inferred: $2*i0+i1+r1$





Conclusions

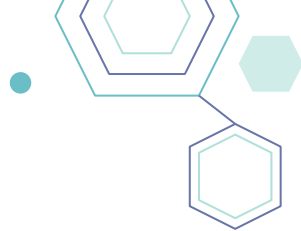
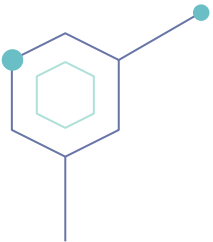
● Evaluation

Based on the known features of the GP tool the results indicates that the evaluation metric is valid.

● Better to implement correlation analysis in the GP tool directly.

As the analysis of correlations allow us to get an indication of the source of the error it would be wise to, as a next step in the process, make use of this in the GP tool itself.





Thank you for listening!

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